Linear equation :- A straight line in the xy plane can be represented by

$$a_1x + a_2y = b$$

$a, a_2, b \to$ real constants
$a_1, a_2 \neq 0$.

in general

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

is a linear equation with 'n' variables.

$a_1 a_2 \dots a_n \to$ Coefficients.

Eg:

$y = mx + c$
$x + 3y = 7$
$x_1 - 2x_2 - 3x_3 + 4x_4 - 9x_5 = 10.$

} examples of linear eqns

$\Rightarrow$

$x^2 + 2x = 5$
$y = \sin x$
$y = e^x$

} not linear eqns $\leftarrow$ $\begin{cases} x = yz \\ x = y^3 + 2z \end{cases}$

Solution of a linear system :-

$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ is the general form of linear eqn.

$\exists$ (Exist) a sequence of numbers $S_1, S_2 \dots S_n$ such that

$a_1S_1 + a_2S_2 + \dots + a_nS_n = b$ then.

The set of all the solutions of the linear eqn. is called general soln/ or solution set.

System of linear eqns : It is a finite set of linear equations, each with the same set of variables. A solution of a linear equation is a vector that is simultaneously a solution of each equation in the system. The solution set of a system of linear eqn is the set of all solution of the system.

Eg:

$2x - y = 3$
$x + y^3 = 5$

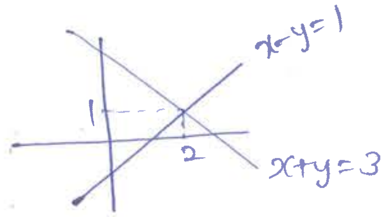
$\to$ $(1, -1)$ does not satisfy eqn(2) so it is not solution set

Ans $\to (2, 1)$

Representation (graphical)

(a) $x-y=1$, $x+y=3$ } soln $[2,1]$

unique solution.

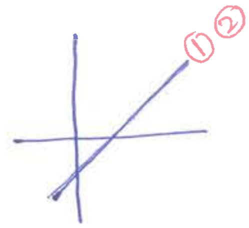


(b) $x-y=2$, $2x-2y=4$ soln $[t, t-2]$

$[t+2, t]$

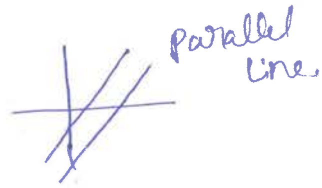
$t \in \mathbb{R}$

$\Rightarrow$ infinite many solutions



Try to visualize with $y=mx+c$

(c) $x-y=1$, $x-y=3$ } no solution



Conclusion:

A system of linear eqns with real coefficients has either

(a) a unique solution (consistent system)

(b) infinite many solutions (")

(c) no solution (inconsistent system)

(*) System of linear equations in matrix form

Let an arbitrary system of 'm' linear equations with 'n' unknowns (variables)

can be written as

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$
$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$
$$\vdots \qquad\qquad \vdots \qquad \vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$

will be:

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix}}_{m \times n} \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_{n \times 1} = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}}_{m \times 1}$$

$\Rightarrow \quad AX = B.$ where A is $m \times n$ matrices with coeffi

B is $m \times 1$ matrices with constant

X is $n \times 1$ variable or, unknown

Eg:

$$x + 2y + 3z = 6$$
$$2x + 5y + 2z = 4$$
$$6x - 3y + z = 2$$

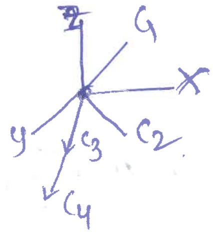
$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 2 \\ 6 & -3 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}$$

$\Rightarrow$ for understanding: (from previous semester) vector notation

we will come to this later!!

it can also written as

$$\underset{C_1}{x\begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}} + \underset{C_2}{y\begin{bmatrix} 2 \\ 5 \\ -3 \end{bmatrix}} + \underset{C_3}{z\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}} = \underset{C_4}{\begin{bmatrix} 6 \\ 4 \\ 2 \end{bmatrix}}$$



* Augmented matrix

→ system of 'm' linear eqns with 'n' unknowns can be represented as

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{bmatrix}$$

is called augmented matrix.

$\rightarrow [A|B]$ or, $[A \vdots B]$,

Eg:

$$x + 2y + 3z = 6$$
$$2x + 5y + 2z = 4$$
$$6x - 3y + z = 2$$

in augmented form $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & 5 & 2 & 4 \\ 6 & -3 & 1 & 2 \end{array}\right]$

*

* methods to solve linear eqn:

1. Direct method
   - Gaussian elimination
   - LU decomposition
   - matrix inversion
   - QR decomposition.
2. Indirect method
   - Iterative method
     - Jacobi method
     - Gauss-Seidel
     - Relaxation method
   - Cramer's rule
3. other methods
   - Eigen value
   - Graphical

* Elementary operations:

In system of linear eqns ;	In Augmented matrix
1. multiply an eqn through by a non-zero constant	1. multiply a row through by a nonzero constant
2. Interchange two equations	2. Interchange two rows
3. Add a multiple of one eqn to another	3. Add a multiple of one row to another row.

Eg.

$$x+2y+2z=9 \rightarrow (1)$$
$$2x+4y-3z=1 \rightarrow (2)$$
$$3x+6y-5z=0 \rightarrow (3)$$

Augmented form.

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 2 & 4 & -3 & 1 \\ 3 & 6 & -5 & 0 \end{array}\right]$$

(2) − 2×(1)

$$x+y+2z = 9 \rightarrow (1)$$
$$2y-7z = -17$$
$$3x+6y-5z = 0$$

$R_2 \rightarrow R_2 - 2R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 3 & 6 & -5 & 0 \end{array}\right]$$

(3) − 3×(1)

$$x+y+2z = 9$$
$$2y-7z = -17 \rightarrow (4)$$
$$3y-11z = -27$$

$R_3 \rightarrow R_3 - 3R_1$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 2 & -7 & -17 \\ 0 & 3 & -11 & -27 \end{array}\right]$$

eq.(4)/2

$$\left[\begin{array}{l} x+y+2z = 9 \\ y-7/2\,z = -17/2 \rightarrow (5) \\ 3y-11z = -27 \rightarrow (6) \end{array}\right]$$

$R_2 \rightarrow R_2/2$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 9 \\ 0 & 1 & -7/2 & -17/2 \\ 0 & 3 & -11 & -27 \end{array}\right]$$

6 − 3×5

$x + y + 2z = 9$

$y - \frac{7}{2}z = -\frac{17}{2}$

$-\frac{z}{2} = -\frac{3}{2} \rightarrow 7$

$R_3 \rightarrow R_3 - 3R_2$

$$\left[\begin{array}{ccc:c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & -\frac{1}{2} & -\frac{3}{2} \end{array}\right]$$

7×−2

$x + y + 2z = 9 \rightarrow 1$

$y - \frac{7}{2}z = -\frac{17}{2} \rightarrow 8$

$z = 3$

$R_3 \rightarrow R_3 \times -2$

$$\left[\begin{array}{ccc:c} 1 & 1 & 2 & 9 \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right]$$

→ upto here gaussian elimin

8×−1 + 1

$x + \frac{11}{2}z = \frac{35}{2} \rightarrow 9$

$y - \frac{7}{2}z = -\frac{17}{2} \rightarrow 8$

$z = 3 \rightarrow 10$

$R_1 \rightarrow R_1 - R_2$

$$\left[\begin{array}{ccc:c} 1 & 0 & \frac{11}{2} & \frac{35}{2} \\ 0 & 1 & -\frac{7}{2} & -\frac{17}{2} \\ 0 & 0 & 1 & 3 \end{array}\right]$$

10 × $-\frac{11}{2}$ and 10 × $\frac{7}{2}$ − 8

$x = 1$

$y = 2$

$z = 3$

$R_1 \rightarrow R_1 - \frac{11}{2}R_3$

$R_2 \rightarrow R_2 + \frac{7}{2}R_3$.

$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array}\right]$$

$$\boxed{\begin{aligned} x &= 1 \\ y &= 2 \\ z &= 3 \end{aligned}}$$

⇒ reduced row-echelon form.

→ Gaussian elimination

1. write the augmented matrix of the system of linear eqns
2. Use elementary row operations to reduce the augmented matrix to row echelon form
3. Using back substitution, solve the equivalent system that corresponding to the row-reduced matrix.

→ ~~row-reduced~~

Reduced row-echelon form. properties

* If a row does not consist entirely of zeros, then the first non zero in the row is one. → leading one.
* If there are any rows that consist entirely of zeros, then they are grouped together at the bottom of the matrix
* In any two successive rows that does not consist entirely of zeros, then the leading 1 in the lower row occurs farther to the right than the leading 1 in the higher row.

Eg row - echelon form

$$\begin{bmatrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 2 & 6 & 0 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

reduced row - echelon form

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix}, \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -2 & 0 & 1 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

* pivot → All the entries ~~above~~ below the pivots are zero (Eliminated)

Incorrect

Eg: $A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & -3 & -5 \\ 0 & 0 & 7 \end{bmatrix}$

⇒ leading non zero element in the row.

* Gaussian elimination: / backsubstitution.

$$AX = B$$

Eg: $-3x_1 + 2x_2 - x_3 = -1$
$6x_1 - 6x_2 + 7x_3 = -7$
$3x_1 - 4x_2 + 4x_3 = -6$

$$\Rightarrow \begin{pmatrix} -3 & 2 & -1 \\ 6 & -6 & 7 \\ 3 & -4 & 4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -1 \\ -7 \\ -6 \end{pmatrix}$$

Augmented matrix $\begin{pmatrix} -3 & 2 & -1 & -1 \\ 6 & -6 & 7 & -7 \\ 3 & -4 & 4 & -6 \end{pmatrix}$

$R_2 \to R_2 + 2R_1$
$R_3 \to R_3 + R_1$ $\sim \begin{pmatrix} -3 & 2 & -1 & -1 \\ 0 & -2 & 5 & -9 \\ 0 & -2 & 3 & -7 \end{pmatrix}$

$R_3 \to R_3 - R_2$ $\sim \begin{pmatrix} -3 & 2 & -1 & -1 \\ 0 & -2 & 5 & -9 \\ 0 & 0 & -2 & 2 \end{pmatrix}$

$\therefore$ $-3x_1 + 2x_2 - x_3 = -1$
$-2x_2 + 5x_3 = -9$
$-2x_3 = 2$

back substitution $x_3 = -1$

$x_2 = \dfrac{-9-5(-1)}{-2} = 2$

$x_1 = 2$

$$\therefore \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}.$$

(*) Reduced row echelon form:

$A = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 5 & 6 & 7 \\ 6 & 7 & 8 & 9 \end{pmatrix}$,

$R_2 \to R_2 - 4R_1$
$R_3 \to R_3 - 6R_1$ $\sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & -3 & -6 & -9 \\ 0 & -5 & -10 & -15 \end{pmatrix}$

$R_3 \to R_3/-5$
$R_2 \to R_2/-3$ $\sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 1 & 2 & 3 \end{pmatrix}$

$R_3 \to R_3 - R_2$ $\sim \begin{pmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$

$R_1 \to R_1 - 2R_2$ $\sim \begin{pmatrix} 1 & 0 & -1 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ rref

O $\to$ pivots $\to$ pivot columns.

prob: $A = \left(\begin{array}{ccc|c} 3 & -7 & -2 & -7 \\ -3 & 5 & 1 & 5 \\ 6 & -4 & 0 & 2 \end{array}\right)$

Soln $\begin{pmatrix} 3 & -7 & -2 & -7 \\ 0 & -2 & -1 & -2 \\ 0 & 0 & -1 & 6 \end{pmatrix}$; $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ -6 \end{pmatrix}$

$\left(\begin{array}{ccc|c} 1 & -2 & 3 & 1 \\ -1 & 3 & -1 & -1 \\ 2 & -5 & -5 & 1 \end{array}\right) = B =$

Soln $\begin{pmatrix} 1 & -2 & 3 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$; $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 8 \\ 2 \\ -1 \end{pmatrix}$

rref : $A \to \begin{pmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & -6 \end{pmatrix}$ $C_1, C_2\ C_3 \to$ pivot col.

rref. $C = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 1 \\ 3 & 6 & 2 \end{pmatrix}$ $C \Rightarrow \begin{pmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ $C_1, C_3 \to$ pivot col.

Compute the inverse

$AA^{-1} = I$

$A = \begin{pmatrix} -3 & 2 & -1 \\ 6 & -6 & 7 \\ 3 & -4 & 4 \end{pmatrix}$;

$\Rightarrow \left(\begin{array}{ccc|ccc} -3 & 2 & -1 & 1 & 0 & 0 \\ 6 & -6 & 7 & 0 & 1 & 0 \\ 3 & -4 & 4 & 0 & 0 & 1 \end{array}\right)$;

$R_2 \to R_2 + 2R_1$
$R_3 \to R_3 + R_1$
$\begin{pmatrix} -3 & 2 & -1 & 1 & 0 & 0 \\ 0 & -2 & 5 & 2 & 1 & 0 \\ 0 & -2 & +3 & 1 & 0 & 1 \end{pmatrix}$

$R_1 \to R_1 + R_2$
$R_3 \to R_3 - R_2$
$\begin{pmatrix} -3 & 0 & 4 & 3 & 1 & 0 \\ 0 & -2 & 5 & 2 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{pmatrix}$

$R_1 \to R_1 + 2R_3$
$R_3 \to$
$\begin{pmatrix} -3 & 0 & 0 & 1 & -1 & 2 \\ 0 & -2 & 5 & 2 & 1 & 0 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{pmatrix}$

$R_2 \to R_2 + \frac{5}{2} R_3$
$\begin{pmatrix} -3 & 0 & 0 & 1 & -1 & 2 \\ 0 & -2 & 0 & -1/2 & -3/2 & +5/2 \\ 0 & 0 & -2 & -1 & -1 & 1 \end{pmatrix}$

$R_1 \to R_1/-3$
$R_2 \to R_2/-2$
$R_3 \to R_3/-2$
$\sim \begin{pmatrix} 1 & 0 & 0 & -1/3 & 1/3 & -2/3 \\ 0 & 1 & 0 & 1/4 & 3/4 & -5/4 \\ 0 & 0 & 1 & 1/2 & 1/2 & -1/2 \end{pmatrix}$

$\therefore A^{-1} = \begin{pmatrix} -1/3 & 1/3 & -2/3 \\ 1/4 & 3/4 & -5/4 \\ 1/2 & 1/2 & -1/2 \end{pmatrix}$

Elementary matrices ; LU decomposition

(Theory)
(Not need)

$$A = \begin{pmatrix} -3 & 2 & -1 \\ 6 & -6 & 7 \\ 3 & -4 & 4 \end{pmatrix}$$

$R_2 \to R_2 + 2R_1$

$$= \begin{pmatrix} -3 & 2 & 1 \\ 0 & -2 & 5 \\ 3 & -4 & 4 \end{pmatrix} \simeq M_1 A \; ; \quad m_1 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & 2 & 1 \\ 0 & -2 & 5 \\ 0 & -2 & 3 \end{pmatrix} = m_2 m_1 A \; ; \quad m_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} -3 & 2 & 1 \\ 0 & -2 & 5 \\ 0 & 0 & -2 \end{pmatrix} = m_3 m_2 m_1 A \; ; \quad m_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

$\Rightarrow m_3 m_2 m_1 A \to U$ upper triangle matrix.

$\therefore m_3 m_2 m_1 A = U$

$\therefore A = m_1^{-1} m_2^{-1} m_3^{-1} U$

$\Rightarrow A = LU.$

$$m_1 = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} ; \quad m_1^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$m_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} ; \quad m_2^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

$$m_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} ; \quad m_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

$$L = m_1^{-1} m_2^{-1} m_3^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix}$$

LU Decomposition:

$Ax = b$

A can be written as LU.

$LUx = b$; let $y = Ux$

$LUx = b$

$Ly = b \Rightarrow$ Solve for y ; forward Substitution.

Solve ; $Ux = y$ for x ; backward Substitution

Eg: Solve for x ; $\exists\ Ax = b$, where $A = \begin{pmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{pmatrix}$; $b = \begin{pmatrix} -3 \\ 3 \\ 2 \end{pmatrix}$

$$A = \begin{pmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{pmatrix} \begin{pmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{pmatrix} = L.U$$

① $LUx = b$

$y = Ux \Rightarrow Ly = b$

$$\begin{aligned} y_1 &= -3 \\ -y_1 + y_2 &= 3 \\ 2y_1 - 5y_2 + y_3 &= 2 \end{aligned}$$

forward substitution

$$y = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 8 \end{pmatrix}$$

② $Ux = y$

$$\begin{aligned} 3x_1 - 7x_2 - 2x_3 &= -3 \\ -2x_2 - x_3 &= 0 \\ -x_3 &= 8 \end{aligned}$$

$\Rightarrow$ backsubstitution

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ -8 \end{pmatrix}.$$

⊛ LU decomposition → are not unique!

It depends on how you choosen L & U:

in general.

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & l_1 & 0 \\ l_{31} & l_{32} & l_3 \end{bmatrix} ; \quad U = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$$

but it also possible to take.

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} ; \quad U = \begin{bmatrix} 1 & U_{12} & U_{13} \\ 0 & 1 & U_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

Eg: $A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 3 \\ -2 & 5 & 5 \end{bmatrix}$ find L & A such that $LU = A$

let $L = \begin{bmatrix} 1 & 0 & 0 \\ L_{21} & 1 & 0 \\ L_{31} & L_{32} & 1 \end{bmatrix}$, $U = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ 0 & U_{22} & U_{23} \\ 0 & 0 & U_{33} \end{bmatrix}$

$$LU = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ L_{21}U_{11} & L_{21}U_{12}+U_{22} & L_{21}U_{13}+U_{23} \\ L_{31}U_{11} & L_{31}U_{12}+L_{32}U_{22} & L_{31}U_{13}+L_{32}U_{23}+U_{33} \end{bmatrix}$$

$A = LU$

$$\therefore \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 3 \\ -2 & 5 & 5 \end{bmatrix} = \begin{bmatrix} U_{11} & U_{12} & U_{13} \\ L_{21}U_{11} & L_{21}U_{12}+U_{22} & L_{21}U_{13}+U_{23} \\ L_{31}U_{11} & L_{31}U_{12}+L_{32}U_{22} & L_{31}U_{13}+L_{32}U_{23}+U_{33} \end{bmatrix}$$

by comparing on ether side

$U_{11} = 2$
$U_{12} = 1$
$U_{13} = 3$

$L_{21}U_{11} = 4$
$L_{21} \cdot 2 = 4$
$L_{21} = 2$

$L_{21}U_{12} + U_{22} = -1$
$2 \cdot 1 + U_{22} = -1$
$U_{22} = -3$

$L_{21}U_{13} + U_{23} = 3$
$2 \cdot 3 + U_{23} = 3$
$U_{23} = -3$

$L_{31}U_{11} = -2$
$L_{31} = -1$

similarly we find other elements

$L_{31}U_{12} + L_{32}U_{22} = 5$
$-1 \cdot 1 + L_{32}(-3) = 5$
$L_{32} = -2$

$L_{31}U_{13} + L_{32}U_{23} + U_{33} = 5$
$-1 \cdot 3 + -2 \cdot -3 + U_{33} = 5$
$U_{33} = 2$

$\therefore L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -2 & 0 \end{bmatrix}$; $U = \begin{bmatrix} 2 & 1 & 3 \\ 0 & -3 & -3 \\ 0 & 0 & 2 \end{bmatrix}$

check $LU = A$.

Other method

$A = \begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 3 \\ -2 & 5 & 5 \end{bmatrix}$

$[R_i \to -kR_j$
$L_{ij}]$

$R_2 \to R_2 - 2R_1$
$R_3 \to R_3 - (-R_1)$

$R_3 \to R_3 + 2R_2 = R_3 - (-2R_2)$

$$\therefore L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -2 & 1 \end{bmatrix}$$

It is easily can find U now → i.e rref is uppertriangle matrix.
~~first row of U is first row of A~~ ;

Q=

$$A = \begin{bmatrix} 3 & 1 & 3 & -4 \\ 6 & 4 & 8 & -10 \\ +3 & 2 & 5 & -1 \\ -9 & 5 & -2 & -4 \end{bmatrix}$$

$R_2 \to R_2 - 2R_1$
$R_3 \to R_3 - R_1$
$R_4 \to R_4 + 3R_1$;

$$\begin{bmatrix} 3 & 1 & 3 & -4 \\ 0 & 2 & 2 & -2 \\ 0 & 1 & 2 & 3 \\ 0 & 8 & 7 & -16 \end{bmatrix}$$

$R_3 \to R_3 - \frac{R_2}{2}$
$R_4 \to R_4 - 4R_2$

$$\begin{bmatrix} 3 & 1 & 3 & -4 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & -1 & -8 \end{bmatrix}$$

$R_4 \to R_4 + R_3$

$$\begin{bmatrix} 3 & 1 & 3 & -4 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & -4 \end{bmatrix} \Rightarrow U$$

For L_{ij} ; The multiplers are

$R_2 \to R_2 - 2R_1$ $\quad L_{21} = 2$
$R_3 \to R_3 - R_1$ $\quad L_{31} = 1$
$R_4 \to R_4 - -3R_1$ $\quad L_{41} = -3$

$R_i - kR_j \to L_{ij}$

and

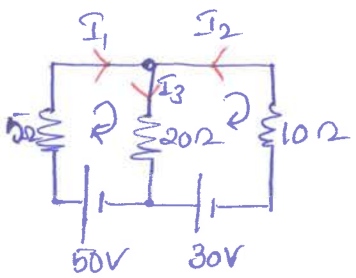
$R_3 \to R_3 - R_2(\frac{1}{2})$ $\quad L_{32} = \frac{1}{2}$
$R_4 \to R_4 - 4R_2$ $\quad L_{42} = 4$
$R_4 \to R_4 - (-R_3)$ $\quad L_{43} = -1$.

$$\therefore L = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 \\ 1 & 1/2 & 1 & 0 \\ -3 & 4 & -1 & 1 \end{bmatrix} ; U = \begin{bmatrix} 3 & 1 & 3 & -4 \\ 0 & 2 & 2 & -2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & -4 \end{bmatrix}$$

7

* (1) ⇒ A voltage drop occurs at a resistors if the direction assigned to the current through the resistor is the same as the direction assigned to the loop, and a voltage rise occurs at a resistor if the direction assigned to the current through the resistor is the opposite to that assigned to the loop.

⇒ A voltage rise occurs at a battery if the direction assigned to the loop is from − to + through the battery and a voltage drop occurs at a battery if the direction assigned to the loop is from + to − through the battery.



$50 - 5I_1 - 20I_3 = 0$

$30 + 20I_3 + 10I_2 = 0$

$-5I_1 + 10I_2 + 30 + 50\,V = 0$

Same direction −

opp dirc. +

$-\!|\!\vdash$ − to +

+ to −

(2) chemical eqn/

Balanced chemical eqn/

$$CH_4 + 2O_2 \to CO_2 + 2H_2O.$$

$$2CH_4 + 4O_2 \to 2CO_2 + 4H_2O.$$

$$x_1(CH_4) + x_2(O_2) \to x_3(CO_2) + x_4(H_2O)$$

	Left		right
Carbon	x_1	=	x_3
Hydrogen	$4x_1$	=	$2x_4$
oxygen	$2x_2$	=	$2x_3 + x_4$

⇒ $\left.\begin{array}{l} x_1 - x_3 = 0 \\ 4x_1 - 2x_4 = 0 \\ 2x_2 - 2x_3 - x_4 = 0 \end{array}\right\}$ System of LE

(3) polynomial

Find a cubic polynomial eqn. passing through $(1,3), (2,-2), (3,-5), (4,0)$.

Clue General form $p = a_0 + a_1x + a_2x^2 + a_3x^3$.

$(1,3) \Rightarrow \quad a_0 + a_1 + a_2 + a_3 = 3.$

$(2,-2) \Rightarrow \quad a_0 + 2a_1 + 4a_2 + 8a_3 = -2$

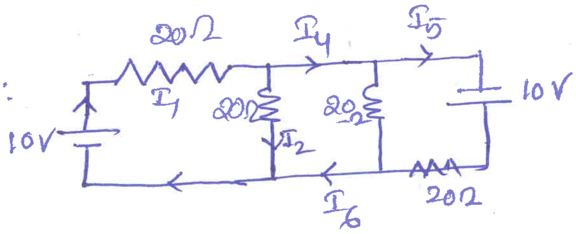
$(3,-5) \Rightarrow \quad a_0 + 3a_1 + 9a_2 + 27a_3 = -5$

$(4,0) \Rightarrow \quad a_0 + 4a_1 + 16a_2 + 64a_3 = 0$

System of LE. using augmented matrix we can find a_0, a_1, a_2, a_3.

Ans $p = 4 + 3x - 5x^2 + x^3$.

prob :



find current in the given electric circuit.